

Zac Freeman
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Experience

Systems Engineer | AutoZone

January 2020 - Present

- Automated a manual provisioning process for OAuth clients with a GitOps system that managed the entire request lifecycle and provided a secure, auditable platform for AI-generated changes
- Led the transformation of a shared Gitlab CI/CD pipeline project from a single-purpose Java deployment tool to the enterprise-standard product for securely deploying any application to any environment
- Built a comprehensive automated test suite for our Gitlab CI/CD pipelines with a focus on easy extensibility to accelerate development without sacrificing reliability as the product surface scaled to ~50 pipeline types, multiple active versions, and over 2000 end-user Gitlab projects
- Performed the duties of contributor, reviewer, and trainer for C++ payment processing software handling \$5B a year across 20,000 devices
- Recovered \$2,000,000 a year in penalties by updating a custom Qt/C++ time clocking software to prompt employees with surveys in compliance with California payroll law
- Rescued a project to replace the employee self-service site for 140,000 employees, releasing a working product on schedule

Software Developer II | Frontdoor

October 2018 - December 2019

- Led the software development for a security initiative, implementing reCAPTCHA for login and registration, session token handling, email verification, and ADFS SSO into a legacy ATG B2B site
- Integrated a 3rd party payment processor to meet new PCI standards for a legacy ATG B2B site
- Designed and built a microservice with Go and Docker to enable B2B clients to place orders in database without interacting with legacy ATG B2B site

Research Assistant | UAH Heliophysics

May 2017 - August 2017

- Created a model of gamma rays in a magnetic trap with the goal of reproducing real world data
- Used C++, CUDA, and the Runge-Kutta method to simulate many parallel gamma ray systems on various NVIDIA GPUs
- Analyzed data produced to discover novel behavior of cosmic rays inside expanding magnetic traps

Research Assistant | FSU Math Department

June 2016 - August 2016

- Used MATLAB and numerical methods for partial differential equations to simulate the flow of liquids and analyze the physical accuracy of the data
- Results from simulations contributed to the paper "Maximally Preserving Finite Difference Schemes for the Allen-Cahn Equation using Operator Splitting", presented at SIAM-SEAS 2017

Education

B.S. in Physics | Florida State University

August 2014 - December 2017

- GPA: 3.889, *Magna Cum Laude*
- Minor: Mathematics

Skills

Languages	Python, Bash, C++, Javascript, $\text{\LaTeX}$, Redstone
Software	Linux, git, Docker, Gitlab CI/CD, Google ADK, Qt